

State-Space Models

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21/10/25

1 Background

- Linear Time-Invariant (LTI) System
- Projection
- Discretisation
- Properties of Matrices

2 State-Space Models (SSM)

- State-Space System
- High-Order Polynomial Projection Operator (HiPPO)
- Structured State-Space System (S4)
- Mamba

3 Summary

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LTI System

Linearity

If the system's input-output relationship is $u(t) \mapsto y(t)$, then for any two inputs $u_1(t)$ and $u_2(t)$ and scalars α, β ,

If $y_1(t) = \mathcal{T}[u_1(t)]$ and $y_2(t) = \mathcal{T}[u_2(t)]$, then $\mathcal{T}[\alpha u_1(t) + \beta u_2(t)] = \alpha y_1(t) + \beta y_2(t)$

Time-Invariance

If the input is shifted in time by τ , then the output is shifted by the same τ ,

$$\mathcal{T}[u(t - \tau)] = y(t - \tau)$$

Connection to Convolution

Denote the system as an operator $\mathcal{T}$ that maps an input signal $u(t)$ to an output signal $y(t)$, i.e., $y(t) = \mathcal{T}[u(t)]$.

Suppose an input is an impulse $u(t) = \delta(t)$, then the output $y(t) = h(t)$ is an impulse response.

Any input $u(t)$ can be decomposed as $u(t) = \int_{\tau} u(\tau) \delta(t - \tau) d\tau$.

If the system is linear, $\mathcal{T}[u(t)] = \int_{\tau} u(\tau) \mathcal{T}[\delta(t - \tau)]$.

If the system is time-invariant, $\mathcal{T}[\delta(t - \tau)] = h(t - \tau)$.

Hence, for the LTI system,

$$y(t) = \mathcal{T}[u(t)] = \int_{\tau} u(\tau) h(t - \tau) = (h * u)(t)$$

In the frequency domain, the convolution becomes multiplication

$$Y(\omega) = H(\omega)U(\omega)$$

where $H(\omega)$ is a transfer function.

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Intuition

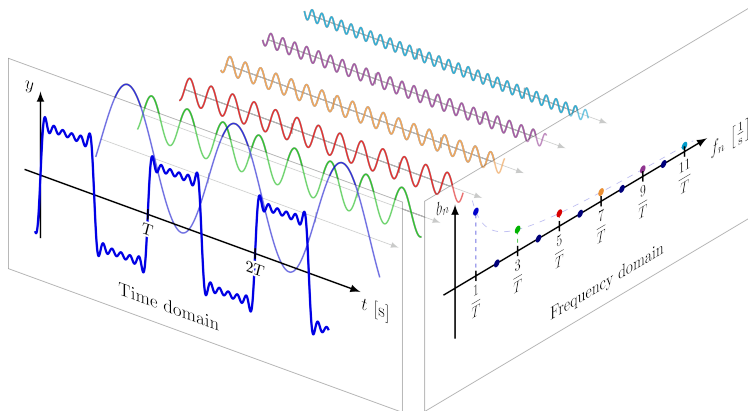


Figure: Fourier transform. Image from <https://dibsmethodsmeetings.github.io/fourier-transforms/>

Basis Function

In function spaces, a set of functions $\{\phi_0(x), \phi_1(x), \dots\}$ are basis functions if a function $f(x)$ can be approximated as a linear combination $f(x) \approx \sum_{n=0}^N c_n \phi_n(x)$ where c_n is a projection coefficient. Basis functions must be linearly independent.

Orthogonality

If an inner product is defined as

$$\langle f, g \rangle = \int f(x)g(x)w(x) dx$$

where $w(x)$ is the weight function, then orthogonal basis functions satisfy

$$\langle \phi_m, \phi_n \rangle = 0 \text{ if } m \neq n$$

Projection coefficient is given by $c_n = \frac{\langle f, \phi_n \rangle}{\langle \phi_n, \phi_n \rangle}$.

Legendre Polynomials

Legendre polynomials $\{P_n(x)\}_{n=0}^{\infty}$ are defined on the interval $[-1, 1]$ as the solutions to Legendre's differential equation

$$\frac{d}{dx}[(1-x^2)\frac{d}{dx}P_n(x)] + n(n+1)P_n(x) = 0$$

A direct definition is

$$P_n(x) = \frac{1}{2^n n!} \frac{d^n}{dx^n} [(x^2 - 1)^n]$$

Legendre polynomials are orthogonal on $[-1, 1]$ w.r.t. the weight function $w(x) = 1$.

- $(1-x^2)P'_n(x) = -nxP_n(x) + nP_{n-1}(x)$
- $(n+1)P_{n+1}(x) = (2n+1)xP_n(x) - nP_{n-1}(x)$

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Discretisation

$\frac{d}{dt}x(t) = Ax(t) + Bu(t)$ can be discretised in the form of $x_{k+1} = \bar{A}x_k + \bar{B}u_k$

Forward Euler (First-Order Approximation)

$$\frac{x_{k+1} - x_k}{\Delta} \approx Ax_k + Bu_k \Rightarrow x_{k+1} = (I + \Delta A)x_k + \Delta Bu_k$$

Backward Euler (First-Order Approximation)

$$\frac{x_{k+1} - x_k}{\Delta} \approx Ax_{k+1} + Bu_{k+1} \Rightarrow x_{k+1} = (I - \Delta A)^{-1}x_k + (I - \Delta A)^{-1}\Delta Bu_{k+1}$$

Bilinear Transform (Second-Order Approximation)

$$\begin{aligned} \frac{x_{k+1} - x_k}{\Delta} &\approx \frac{1}{2}(Ax_k + Ax_{k+1} + Bu_k + Bu_{k+1}) \\ \Rightarrow x_{k+1} &= (I - \frac{\Delta}{2}A)^{-1}(I + \frac{\Delta}{2}A)x_k + (I - \frac{\Delta}{2}A)^{-1}\frac{\Delta}{2}B(u_k + u_{k+1}) \end{aligned}$$

Discretisation

Zero-Order Hold (ZOH)

For any interval, the exact solution to ODE is given by

$$x(t) = e^{A(t-t_0)}x(t_0) + \int_{s=t_0}^t e^{A(t-s)}Bu(s) ds$$

Assume $u(s) = u_k$ is piecewise constant on each interval $s \in [k\Delta, (k+1)\Delta]$ for a learned step size Δ .

Set $t_0 = k\Delta$, $t = (k+1)\Delta$, and $\tau = t - s$, then

$$x_{k+1} = e^{A\Delta}x_k + \left(\int_{\tau=0}^{\Delta} e^{A\tau} d\tau\right)Bu_k$$

If A is invertible, then $\frac{d}{d\tau}(e^{A\tau}) = Ae^{A\tau} \iff e^{A\tau} = A^{-1}\frac{d}{d\tau}e^{A\tau}$ and $\int_{\tau=0}^{\Delta} e^{A\tau} d\tau = A^{-1}[e^{A\tau}]_{\tau=0}^{\Delta} = A^{-1}(e^{A\Delta} - I)$.

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Definition

Symmetry

$A \in \mathbb{R}^{N \times N}$ is symmetric if $A^T = A$.

Normality

$A \in \mathbb{C}^{N \times N}$ is normal if $A^*A = AA^*$.

Orthogonality

$V \in \mathbb{R}^{N \times N}$ is orthogonal if $V^T V = I \iff V^{-1} = V^T$.

Unitarity

$U \in \mathbb{C}^{N \times N}$ is unitary if $U^*U = I \iff U^{-1} = U^*$.

Definition

Diagonality

A is orthogonally diagonalisable if A is similar to a diagonal matrix D w/ an orthogonal matrix V , i.e. $A = VDV^T$.

A is unitarily diagonalisable if A is similar to a diagonal matrix D w/ an unitary matrix U , i.e. $A = UDU^*$.

Singular Value

$\forall A \in \mathbb{C}^{M \times N}$, singular values are defined as $\sigma_i(A) = \sqrt{\lambda_i(A^*A)} \geq 0$.

Spectral Norm

$\|A\|_2 = \sigma_{\max}(A)$, i.e., the largest singular value tells the maximum amplification A can apply to any unit vector.

Theorem

Theorem 1

$A \in \mathbb{R}^{N \times N}$ is symmetric. $\iff A$ is orthogonally diagonalisable.

Theorem 2

$A \in \mathbb{C}^{N \times N}$ is normal. $\iff A$ is unitarily diagonalisable.

Woodbury Matrix Identity

$(A + UCV)^{-1} = A^{-1} + A^{-1}U(C^{-1} + VA^{-1}U)^{-1}VA^{-1}$ where $A \in \mathbb{C}^{N \times N}$, $U \in \mathbb{C}^{N \times k}$, $C \in \mathbb{C}^{k \times k}$, $V \in \mathbb{C}^{k \times N}$.

Mode & Coupling

If A is diagonalisable, then $A = V\Lambda V^{-1}$ where $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_N)$ and $V = [v_1 \cdots v_N]$

Each pair (λ_i, v_i) defines a mode:

- Decay rate
 - $\Re(\lambda_i) < 0$: Decaying mode $\rightarrow$ Memory fades exponentially
 - $\Re(\lambda_i) = 0$: Persistent mode $\rightarrow$ Infinite memory (Oscillation)
 - $\Re(\lambda_i) > 0$: Growing mode $\rightarrow$ Unstable (Exploding memory)
- Oscillation
 - $\Im(\lambda_i) = 0$: No oscillation
 - $\Im(\lambda_i) \neq 0$: Sinusoidal oscillation

Each eigenvector defines a mode direction, and its eigenvalue dictates how that mode evolves. Also, $e^{At} = Ve^{\Lambda t}V^{-1} = V\text{diag}(e^{\lambda_i t})|_{i=1}^n V^{-1}$.

Coupling is modes influencing each other's evolution:

- Diagonal: None
- Diagonal + Low-rank: Light global coupling
- Triangular / Dense: Strong hierarchical coupling

Transient Amplification

Given a homogeneous linear ODE $\frac{d}{dt}x(t) = Ax(t)$ w/ solution $x(t) = e^{At}x(0)$, if all eigenvalues of A have negative real parts, then asymptotically $x(t) \rightarrow 0$.

When A is normal, the eigenvectors form an orthogonal unitary basis, and energy evolves independently in each mode

When A is non-normal, the directions overlap — energy in one mode can project onto another, causing transient amplification, i.e., $\|x(t)\|_2$ increases before decaying.

The amplification (gain) at time t is $G(t) = \frac{\|x(t)\|}{\|x(0)\|}$.

The worst-case amplification is given by $G_{\max}(t) = \sup_{t \geq 0} \|e^{At}\|_2$.

- $G_{\max} > 1 \iff A$ is non-normal.
- $G_{\max} = 1 \iff A$ is normal w/ all $\Re(\lambda_i) \leq 0$.

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State-Space System

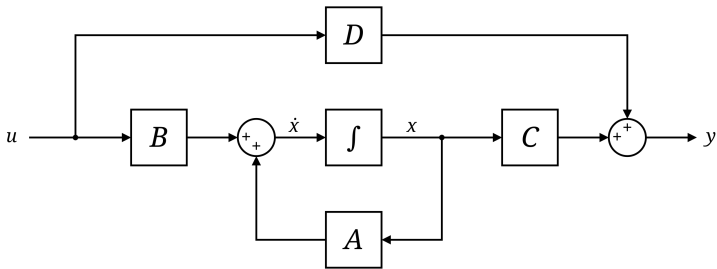


Figure: SSM. Image from

https://www.wikipedia.com/en/articles/State-space_representation

Continuous-Time Linear State-Space System

The system is represented as

$$\begin{cases} \frac{d}{dt}x(t) = Ax(t) + Bu(t) \\ y(t) = Cx(t) + Du(t) \end{cases}$$

where

- $x(t)$: State vector (Hidden memory of the system)
- $u(t)$: Input signal (External sequence coming in)
- $y(t)$: Output signal (Observed sequence)
- A, B, C, D : Matrices

Feed-through part $Du(t)$ is the residual connection and can be omitted for the simplicity of the future computation.

The system is LTI iff there are no non-linear functions, and when A, B, C, D are constant matrices which do not depend on t .

Discrete-Time Linear State-Space System

After discretisation, the system turns into

$$\begin{cases} x_{t+1} = \bar{A}x_t + \bar{B}u_t \\ y_t = \bar{C}x_t + \bar{D}u_t \end{cases}$$

where the first equation is the recurrent network.

Unrolling the recurrence, given the entire input history $\{u_0, \dots, u_t\}$,

$x_t = \bar{A}^t x_0 + \sum_{k=0}^{t-1} \bar{A}^k \bar{B} u_{t-1-k}$ where the first term is the homogeneous term which depends on the initial state, and the second term is the particular solution that represents the convolution w/ past inputs.

Setting $x_0 = 0$, $y_t = \bar{C}(\sum_{k=0}^{t-1} \bar{A}^k \bar{B} u_{t-1-k}) = \sum_{k=0}^{t-1} K_k u_{t-1-k}$ where $K_k = \bar{C} \bar{A}^k \bar{B}$ is the kernel.

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HiPPO: Polynomial Projection of History

Suppose the past input function is $u(\tau)$ for $\tau \in [0, t]$.

HiPPO approximates the function using a set of basis functions $p_n(\cdot)$:

$$u(\tau) \approx \sum_{n=0}^{N-1} x_n(t) p_n\left(\frac{\tau}{t}\right)$$

where $p_n(\cdot)$ are orthogonal polynomials and $x_n(t)$ are projection coefficients.

$x(t)$ is the memory state — the best degree- $(N - 1)$ polynomial approximation of the full history up to time t .

Coefficients are defined by the projection

$$x_n(t) = \int_0^t p_n\left(\frac{\tau}{t}\right) u(\tau) w_t(\tau) d\tau$$

HiPPO: Dynamics via Expanding Integration Window

Suppose time increases slightly from $t \rightarrow t + \Delta t$, then

- ① The integration window expands w/ new data $u(t + \Delta t)$
- ② The basis changes from $p_n(\frac{\tau}{t}) \rightarrow p_n(\frac{\tau}{t+\Delta t})$

This adjustment defines a DE for $x(t)$.

Leibniz rule says

$$\frac{d}{dt} \int_0^t f(\tau, t) d\tau = f(t, t) + \int_0^t \frac{\partial f}{\partial t}(\tau, t) d\tau$$

Let $f(\tau, t) = p_n(\frac{\tau}{t})u(\tau)w_t(\tau)$, then

$$\frac{d}{dt} x_n(t) = p_n(1)u(t)w_t(t) + \int_{\tau=0}^t \frac{\partial}{\partial t} p_n(\frac{\tau}{t})u(\tau)w_t(\tau) d\tau$$

The first term comes from the new data entering the window, and the second term arises because the basis itself changes w/ t .

HiPPO: Deriving the State-Space Equation

For the uniform weight $w_t(\tau) = \frac{1}{t}$,

$$\frac{\partial}{\partial t}(p_n(\frac{\tau}{t})w_t(\tau)) = -\frac{1}{t^2}(\tau p_n'(\frac{\tau}{t}) + p_n(\frac{\tau}{t}))$$

Let $\tau = ts$ then

$$\begin{aligned} \int_{\tau=0}^t \frac{\partial}{\partial t} p_n(\frac{\tau}{t}) u(\tau) w_t(\tau) d\tau &= -\frac{1}{t^2} \int_{\tau=0}^t (\tau p_n'(\frac{\tau}{t}) + p_n(\frac{\tau}{t})) u(\tau) d\tau \\ &= -\frac{1}{t} \int_{s=0}^1 (s p_n'(s) + p_n(s)) u(ts) ds \end{aligned}$$

As $u(ts) \approx \sum_{m=0}^{N-1} x_m(t) p_m(s)$,

$$\frac{d}{dt} x_n(t) = \frac{1}{t} \sum_{m=0}^{N-1} A_{nm} x_m(t) + \frac{1}{t} B_n u(t) = A(t) x(t) + B(t) u(t)$$

where $A_{nm} = -\int_{s=0}^1 (s p_n'(s) + p_n(s)) p_m(s) ds$ and $B_n = p_n(1)$.

HiPPO: Lower-Triangular Structure and Hierarchical Memory

$A_{nm} = -\int_{s=0}^1 (sp'_n(s) + p_n(s))p_m(s) ds = -\langle sp'_n(s) + p_n(s), p_m(s) \rangle = -c_{nm}\|p_m\|^2$ tells that each time derivative of the coefficient $x_n(t)$ depends on a weighted projection of the derivative of basis function p_n onto all the other basis p_m .

For any polynomial basis $p_n(s)$ of degree n , the derivative $p'_n(s)$ can be represented as a linear combination of the lower-order polynomials $p'_n(s) = \sum_{m=0}^{n-1} c_{nm}p_m(s)$.

Since the derivative of the n -th polynomial has no contribution from higher-degree polynomials $\{p_{n+1}, \dots\}$, $A_{nm} = 0 \quad \forall m > n$, i.e. A is lower-triangular.

As A is lower-triangular, the coefficients of $x_n(t)$ depends on $\{x_n, \dots, x_0\}$ leading to hierarchical memory.

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S4: Motivation

A needs to be chosen carefully s.t. it is stable ($\Re(\lambda_i) < 0$) and diagonalisable (to compute $e^{A\Lambda}$ easily).

S4 chooses a specific structured form $A = \Lambda + PQ^*$ where $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_N)$, a normal diagonal matrix w/ complex eigenvalues where $\Re(\lambda_i) < 0$, and $P, Q \in \mathbb{C}^{N \times r}$, a low-rank correction w/ $r = 1$ or 2 .

Diagonal part gives independent modes w/ stable exponential decay, and the low-rank part adds mode coupling.

S4 is LTI, i.e., A, B, C are fixed.

S4: Fast Kernel Computation

The system of equations in the time-domain can be transformed into that of frequency-domain

$$\begin{cases} x_{t+1} = \bar{A}x_t + \bar{B}u_t \\ y_t = \bar{C}x_t \end{cases} \longleftrightarrow \begin{cases} zX(z) = \bar{A}X(z) + \bar{B}U(z) \\ Y(z) = \bar{C}X(z) = K(z)U(z) \end{cases}$$

where $K(z) = \bar{C}(I - \bar{A}z^{-1})^{-1}\bar{B}$.

Naïve kernel coefficients $K_k = \bar{C}\bar{A}^k\bar{B}$ are slow to compute, but evaluating $K(z)$ at many frequency points $z = e^{i\omega}$ is cheap because $A = \Lambda + PQ^*$ has a structured form.

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Mamba: Motivation

In attention, the update depends on the content of tokens: the model dynamically decides which past tokens are relevant.

In S4, the updates A, B, C are fixed linear dynamics, independent of what the current input is.

Mamba tries to make the dynamics input-dependent by redefining the SSM.

$$\begin{cases} x_{t+1} = A(u_t)x_t + B(u_t)u_t = A_t x_t + B_t u_t \\ y_t = Cx_t \end{cases}$$

Matrices A, B are no longer constant, but the functions of the current input u_t . Each token determines how strongly the state decays or updates, so the model becomes selective.

Mamba: Gate

The selectivity is implemented through gates — learned projections of the input controlling the state coefficients.

For each token $u_t \in \mathbb{R}^d$

- Gate signal: $g_t = \sigma(W_g u_t)$
- Input projection: $\tilde{u}_t = W_u u_t$
- State retention: $\bar{A}_t = \text{diag}(\alpha) \odot g_t$
- Input injection: $\bar{B}_t = \text{diag}(\beta) \odot (1 - g_t)$

where σ is a non-linear function and $\alpha, \beta \in \mathbb{R}^N$ are the learned per-channel base coefficients.

g_t dynamically decides how much to keep from the past versus how much to inject from the new input.

The gate operates multiplicatively on linear terms — introducing nonlinearity globally while preserving linearity locally.

Mamba: Selective Scan

By induction $x_{t+1} = A_t x_t + B_t u_t$ can be unrolled as

$$x_t = \left(\prod_{j=0}^{t-1} A_j \right) x_0 + \sum_{k=0}^{t-1} \left(\prod_{j=k+1}^{t-1} A_j \right) B_k u_k$$

Define $\mathcal{S}_t = (A_t, b_t)$ where $b_t = B_t u_t$.

Also define the operator $\otimes$ on adjacent steps $(A_2, b_2) \otimes (A_1, b_1) = (A_2 A_1, b_2 + A_2 b_1)$.

This operation is associative, i.e., $(\mathcal{S}_3 \otimes \mathcal{S}_2) \otimes \mathcal{S}_1 = \mathcal{S}_3 \otimes (\mathcal{S}_2 \otimes \mathcal{S}_1)$.

Hence, a parallel prefix scan can be used.

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Model Cost

Model	Training Cost	Memory	Inference Cost
Transformer	$O(T^2 d)$	$O(T^2)$	$O(T^2)$
Linear Transformer	$O(T d^2)$	$O(T d)$	$O(T d)$
CNN	$O(T k d^2)$	$O(T d)$	$O(k d^2)$ per step
RNN	$O(T d^2)$	$O(T d)$	$O(d^2)$ per step
Linear RNN	$O(T d^2)$	$O(T d)$	$O(d^2)$ per step
S4	$\tilde{O}(T d)$	$O(T d)$	$O(d^2)$ per step
Mamba	$\sim O(T d)$	$O(T d)$	$O(d^2)$ per step

Table: Comparison of training cost, memory, and inference cost across different sequence models. Here T = sequence length, d = hidden dimension, k = kernel size.

Properties 1

Model	Key Equation	Type	Parameterization of A	Complexity
SSM	$\dot{x} = Ax + Bu$	LTI	Arbitrary	$O(TN^2)$
HiPPO	$\dot{x} = A(t)x + B(t)u$	LTV	Lower-triangular, time-varying	$O(TN^2)$
S4	$\dot{x} = (\Lambda + PQ^*)x + Bu$	LTI	Diagonal + Low-rank	$O(TN + N \log N)$
Mamba	$x_{t+1} = A(u_t)x_t + B(u_t)u_t$	LTV	Data-dependent A, B	$O(TN)$

Table: Structural properties of SSM, HiPPO, S4, and Mamba.

Properties 2

Model	Memory Mechanism	Stability	Interpretation
SSM	Exponential decay	Depends on $\Re(\lambda_i)$	Classical control system
HiPPO	Polynomial projection	Stable (structured A)	Optimal memory of continuous input
S4	Long-range stable memory	Stable ($\Re(\lambda_i) < 0$)	Efficient structured SSM
Mamba	Selective (input-adaptive)	Stable (selective control)	Adaptive, hardware-aware SSM

Table: Dynamic and interpretive properties of SSM, HiPPO, S4, and Mamba.